Design Challenge WS23/24

Exercise: Spacecraft Propulsion and Design Challenge

Chair of Space Propulsion and Mobility



Image Source (JPL Hedgehog): https://www-robotics.jpl.nasa.gov/how-we-do-it/systems/hedgehog/

General Announcements

Annoucement

Next week the consultation-hour will be purely online.

General Announcement

Exercise & Consultation-Hour on 01/12/2023 will only be online!

Exercise & Additional Topics

The Report

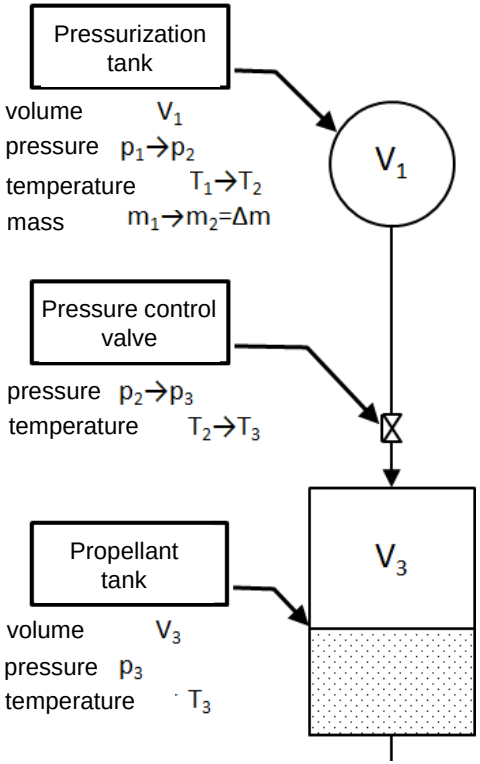
The **Technical Design** part of the report should include the following:

- Description of the propulsion system architecture and design decisions
- P&ID / Schematic of the propulsions system ✓
- List of Equipment (of complete propulsion system) ✓
- Spacecraft Mass and Power budget ✓
- Δv- and Propellant Budget ✓
- Fill level-, pressure and temperature evolution in propellant and pressurant tank (if applicable)
- Technology Needs
- (Thruster Configuration & Control Authority)

ESA CDF Reports can serve as a reference here.

Pressurant Tank Sizing

Design (1/2)



Required pressurization gas mass:

$$\Delta m = V_3 \cdot \rho_3 = \frac{V_3 \cdot p_3 \cdot M_{p.gas}}{\Re \cdot T_3}$$

$$\Delta m = m_1 - m_2 = V_1 \cdot (\rho_1 - \rho_2)$$

Required gas tank volume:

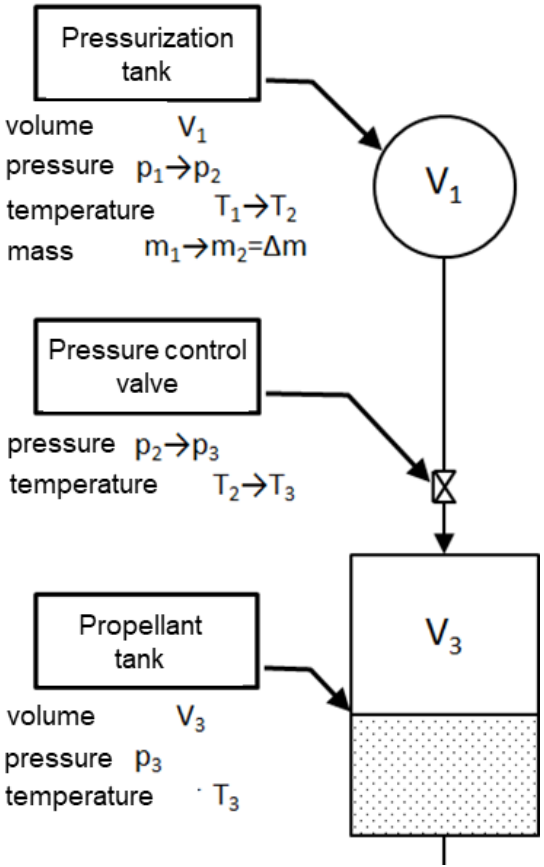
$$V_1 = \frac{V_3 \cdot p_3 \cdot T_2}{p_2 \cdot T_3} \left[\left(\frac{p_1}{p_2} \right)^{\frac{1}{n}} - 1 \right]^{-1}$$

with $$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}}$$

n = polytropic exponent

larger volumes V_3 only reachable with either increasing pressure ratio p_1/p_2 or gas volume V_1

Design (1/2)



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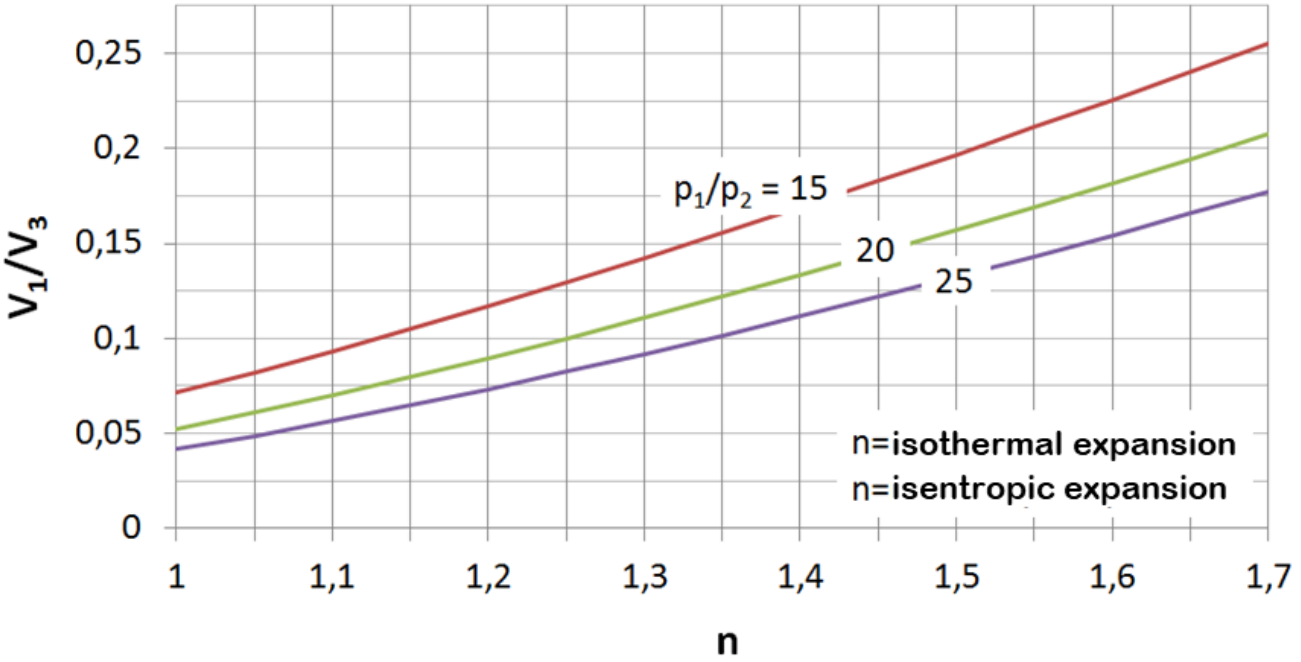
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n = polytropic exponent

larger volumes V_3 only reachable with either increasing pressure ratio p_1/p_2 or gas volume V_1

Design (2/2)



Operation with mit He (κ = 1,67) as pressurization gas	**n**
Strong intermittent operation at very small thrust levels (i.e. attitude control)	1,0
Continuous operation at moderate thrust levels (i.e. upper stage)	1,1 bis 1,2

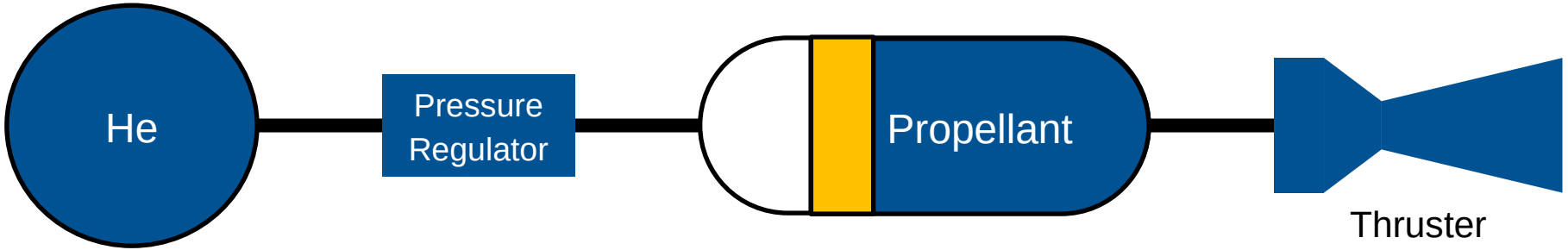
Pressurant Tank Sizing

For large maneuvres a lot of pressurant gas is depleted in one step, causing significant changes in temperature and pressure. Thus it is quite difficult to just use a certain polytropic exponent. In such cases the pressure and temperature evolution has to be calculated step wise. The NESC Academy is providing a good tutorial on that subject:

https://nescacademy.nasa.gov/video/2a5828243b7649ad8283d34e391e76741d

Stepwise Calculation of p & T evolution

Simplification of a pressure regulated system



Stepwise Calculation of p & T evolution

Step 1:

- Propellant mass increment leaves the propellant tank



$$\Delta m_{prop,out} = \dot{m}_{prop} \cdot t_{step}$$

$$m_{prop,f} = m_{prop,i} - \dot{m}_{prop} \cdot t_{step}$$

Stepwise Calculation of p & T evolution



Step 2:

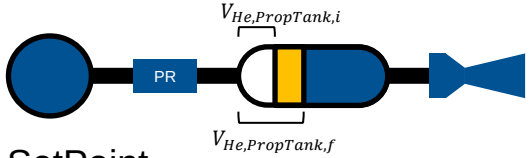
- Volume occupied by propellant decreases
- Volume occupied by Helium increases
- Propellant tank pressure decreases

$$V_{prop,f} = \frac{m_{prop,f}}{\rho_{prop}}$$

$$V_{He,PropTank,f} = V_{Prop\ Tanks} - V_{prop,f}$$

$$p_{PropTank,intermediate} = \frac{m_{He,PropTank,i} \cdot R \cdot T_{He,PropTank,i}}{V_{He,PropTank,f}}$$

Stepwise Calculation of p & T evolution



Step 3:

- Pressure Regulator opens if p_PropTank,intermediate < p_SetPoint
- New Helium flows into propellant tank until p_PropTank ,intermediate =p_SetPoint

$$\Delta V_{He} = V_{He,PropTank,f} - V_{He,PropTank,i}$$

$$V_{He,PropTank,i} = \frac{m_{He,PropTank,i} \cdot R \cdot T_{He,PropTank,i}}{p_{PR,Setpoint}}$$

➡ Determine He mass required to fill this volume

$$\Delta m_{He,PropTank} = \frac{p_{PR,Setpoint} \cdot V_{He}}{R \cdot T_{He,PR\ out}}$$

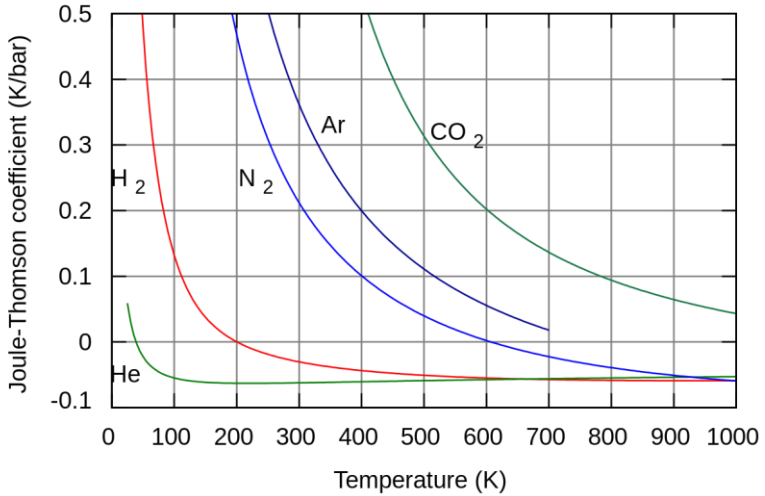
Stepwise Calculation of p & T evolution

Step 3:

- Joule-Thomson effect has to be considered here!

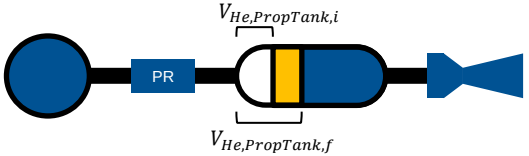


$$T_{He,PR\ out} = T_1 - \mu_{JT}(p_1 - p_2)$$



Joule-Thomson coefficient can be obtained from NIST database or calculated

Stepwise Calculation of p & T evolution



Step 3:

- Pressure Regulator opens if p_PropTank < p_SetPoint
- New Helium flows into propellant tank until p_PropTank=p_SetPoint

$$\Delta m_{He,PropTank} = \frac{p_{PR,Setpoint} \cdot \Delta V_{He}}{R \cdot T_{He,PR\ out}}$$

$$m_{He,PropTank,f} = m_{He,PropTank,i} + \Delta m_{He,PropTank}$$

Stepwise Calculation of p & T evolution

Step 4:

- Helium Temperature in propellant tank is balancing
- New propellant tank pressure establishes



$$T_{He,PropTank,f} = \frac{T_{He,PR\ out} \cdot \Delta m_{He,PropTank} + T_{He,PropTank,i} \cdot m_{He,PropTank,i}}{\Delta m_{He,PropTank} + m_{He,PropTank,i}}$$

$$p_{PropTank,f} = \frac{m_{He,PropTank,f} \cdot R \cdot T_{He,PropTank,f}}{V_{He,PropTank,f}}$$

Stepwise Calculation of p & T evolution



Step 5:

- New Helium mass in Helium tank can be obtained
- New Helium pressure and temperature in Helium tank can be obtained (adiabatic process)

$$m_{He,HeTank,f} = m_{He,HeTank,i} - \Delta m_{He,PropTank}$$

Using the specific volume as intermediate step to determine final T & p

$$v_1 = \frac{V_{HeTank}}{m_{He,HeTank,i}} \qquad v_2 = \frac{V_{HeTank}}{m_{He,HeTank,f}}$$

$$T_{He,HeTank,f} = \frac{T_{He,HeTank,i} \cdot {v_1}^{k-1}}{{v_2}^{\kappa-1}}$$

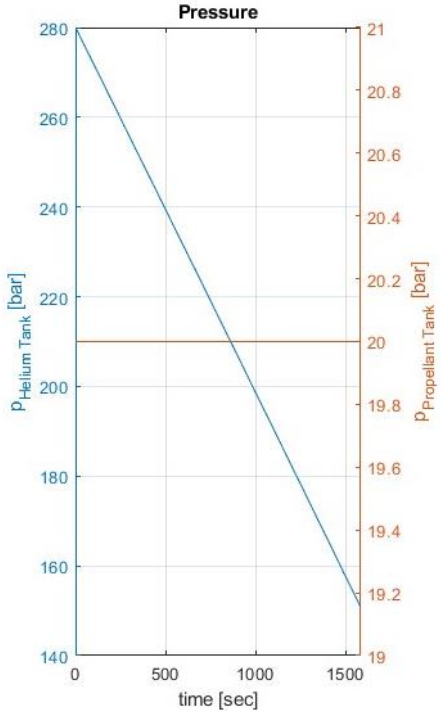
$$p_{He,HeTank,f} = \frac{p_{He,HeTank,i} \cdot {v_1}^{k}}{{v_2}^{\kappa}}$$

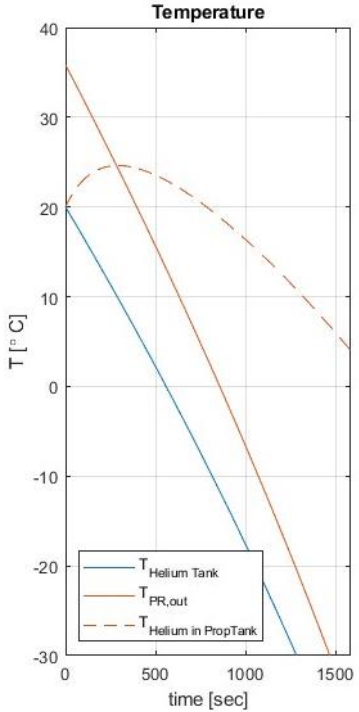
κ = Heat capacity ratio

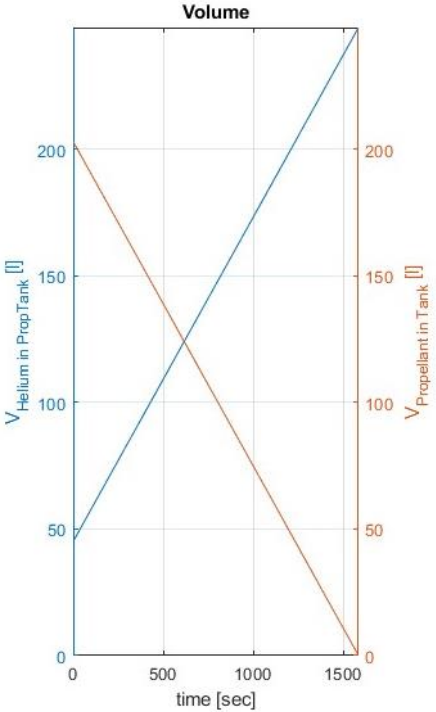
Stepwise Calculation of p & T evolution

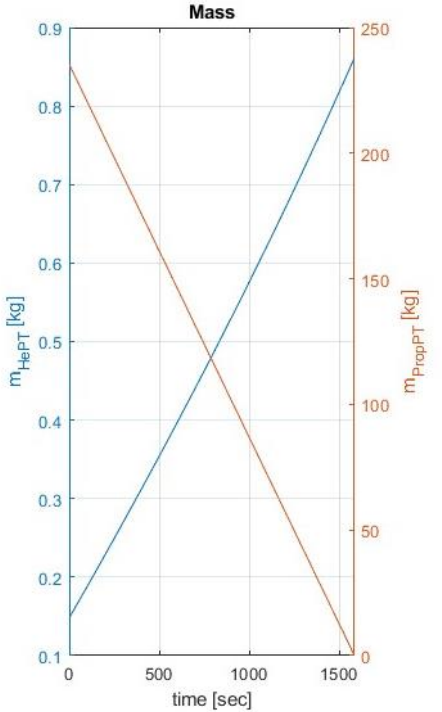
Example











Stepwise Calculation of p & T evolution

Example

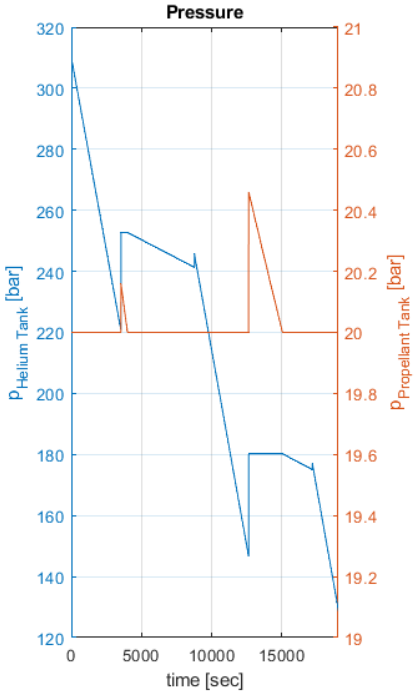
Delta-v [m/s]	Used Thruster #1	Used Thruster #2	Result. Isp [s]	mp_total [kg/s]	m_0 [kg]	m_f [kg]	m_prop_used [kg]
590	Leros2b	S10_18	316.01	0.148	3000.0	2475.2	524.8
85	S10_18	None	292.00	0.014	2475.2	2402.0	73.1
835	Leros2b	S10_18	316.01	0.148	2402.0	1829.7	572.3
100	S10_18	None	292.00	0.014	1829.7	1766.3	63.4
560	Leros2b	S10_18	316.01	0.148	1766.3	1471.6	294.7

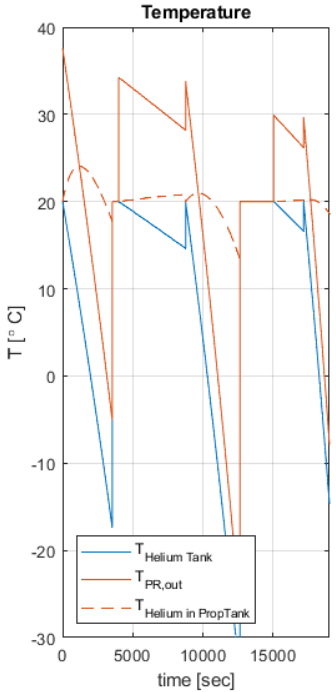
V_MMH = V_MON = 800 [L]

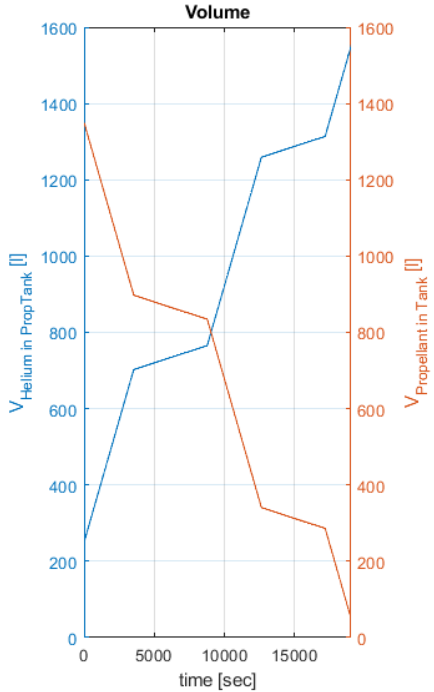
V_Helium = 160 [L], p_HeliumTank_0 = 310 [bar]

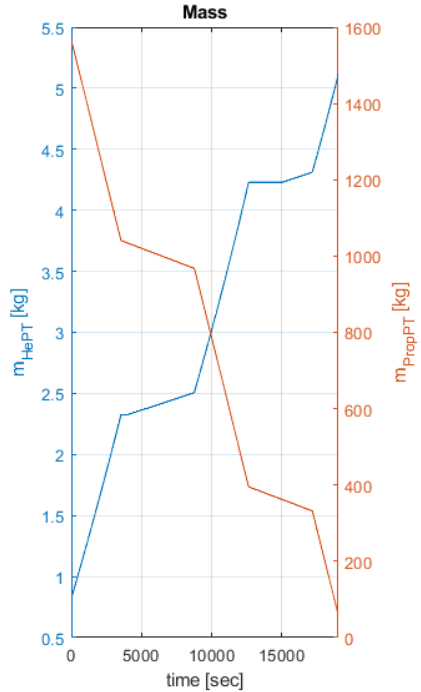
Stepwise Calculation of p & T evolution

Example



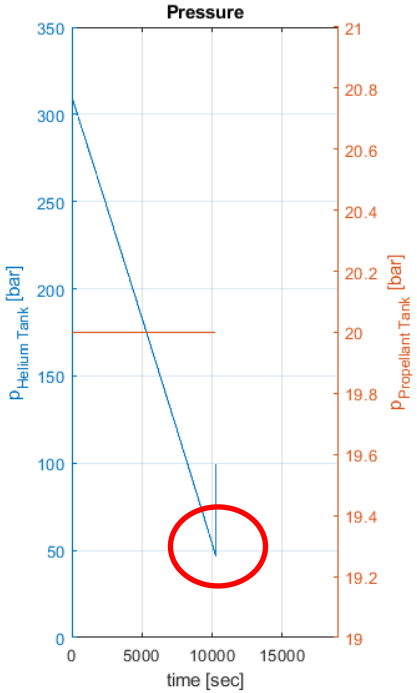


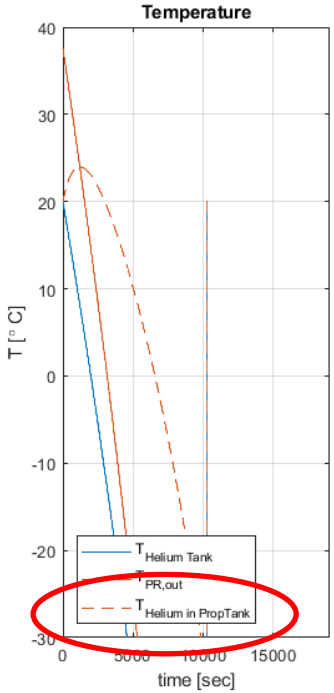


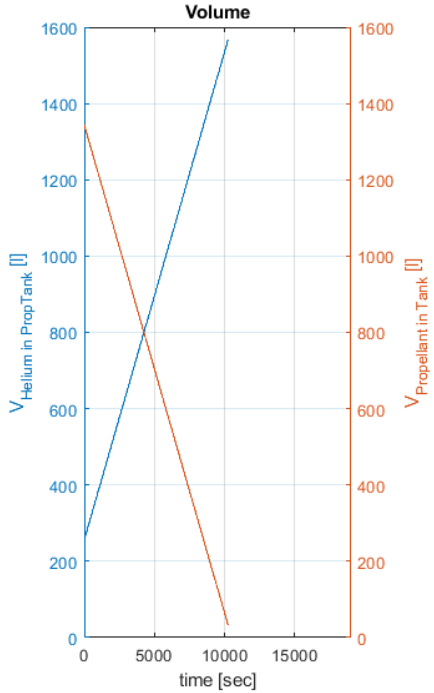


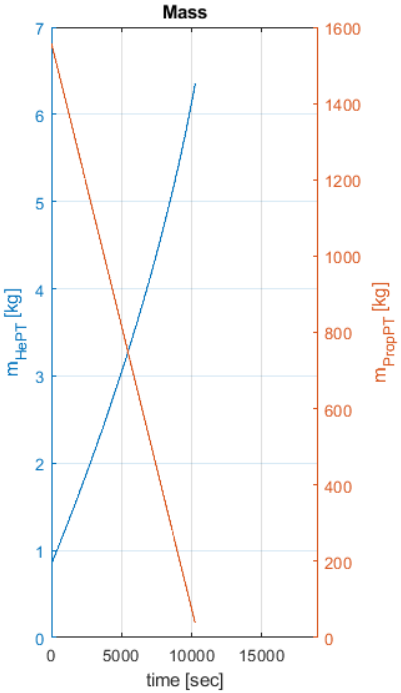
Stepwise Calculation of p & T evolution

Example 2: same total Δv applied at once









Pressure in Helium tank drops much further and
Temperature in propellant tank might freeze the propellant

Technology Needs

Technology Needs

Highlights all technologies that will need to be developed or matured to make the mission feasible

Technology Needs						
*	Equipment Name & Text Reference	Technology	Supplier (Country)	TRL	Funded by	Additional Information
	Electronic Pressure regulator	Pressure regulator using the high proportional valve for MON/MMH	Nammo (UK)	5		
	High Proportional Valve	This valve can be used for throttling purposes in MON/MMH/Hydrazine	Nammo (UK)	5		
	Normally Open Shape Memory Alloy valve	This valve could be used as a replacement for normally open pyrovalves without the lifetime limitation	ArianeGroup (Germany)	1-3		

Source: ESA CDF Study Report Ice Giants